

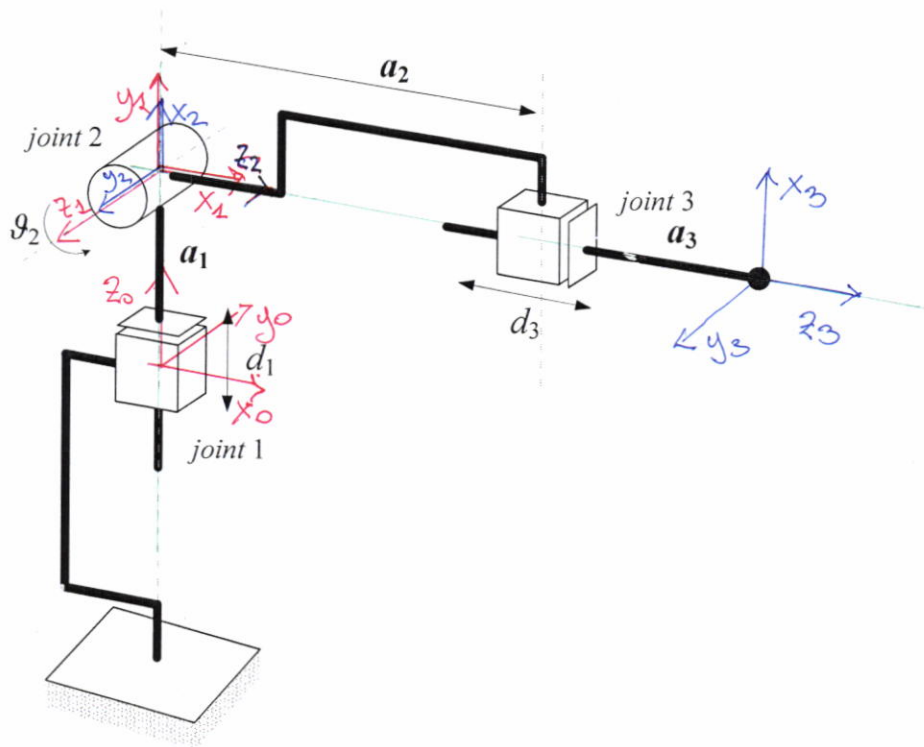
Quesito A

Sia assegnato il manipolatore in figura, con lunghezze dei bracci $a_1 = a_3 = 2$, $a_2 = 1$.

1. Si fissino le terne solidali con ciascun braccio seguendo la convenzione di Denavit-Hartenberg, giustificando adeguatamente la scelta;
2. Si calcoli la funzione cinematica diretta;
3. Si calcoli lo Jacobiano geometrico;
4. Si calcolino posizione e orientamento delle terne solidali con ciascun braccio, rispetto alla terna base, quando il vettore delle variabili di giunto assume il valore $q = [1 \quad \pi/2 \quad 0.5]^T$.
5. A partire dalla configurazione al punto precedente, si definisca una traiettoria per le variabili di giunto in modo tale che l'origine della terna dell'organo terminale descriva nello spazio un arco di circonferenza di ampiezza $\pi/2$ in un intervallo di tempo di 2 secondi. Nello specifico, si assuma che velocità e accelerazioni iniziali e finali siano nulle, e si adottino leggi temporali polinomiali per le variabili di giunto.
6. Con riferimento al punto 5., si individuino posizione e orientamento della terna dell'organo terminale nella posizione finale attraverso la funzione cinematica diretta. Si esprima quindi l'orientamento della terna dell'organo terminale in coordinate ZYZ di Eulero.
7. Con riferimento al punto 5. si determini la velocità di ciascuna terna rispetto al riferimento fisso nell'istante $t = 1$.

Quesito B

Si discuta del controllo PD con compensazione di gravità nello spazio dei giunti di un manipolatore a n giunti. In particolare, si definiscano chiaramente obiettivi e specifiche, si illustri nel dettaglio la metodologia di progetto, e si disegni lo schema di controllo.



A.1 (setting link frames)

①

Setting the link frames according D-H rules

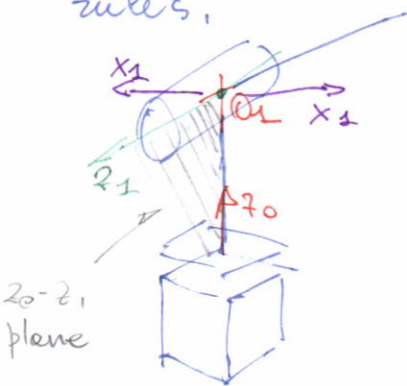
- Identify joint axes and set z_i axes accordingly

z_0, z_1, z_2 are uniquely defined, z_3 is not (joint # does not exist)

For z_3 it's convenient to set the same direction as z_2 , to obtain $a_3=0$ and $\alpha_3=0$. It also holds for $x_3 \parallel x_2$ and $y_3 \parallel y_2$.

- As for O_0 (origin of base frame ϕ), it must be along z_0 direction. Since it's a fixed point (frame ϕ is a fixed frame), it is a good choice to put it in the centre of joint 1. In this case, when $d_1=0$, O_0 and O_1 (origin of frame #1) coincide.

- z_1 direction is determined by joint #2 axis, while O_1 is set by the intersection of z_0 and z_1 directions. Therefore, O_1 must be in the centre of joint #1, according to D-H rules.

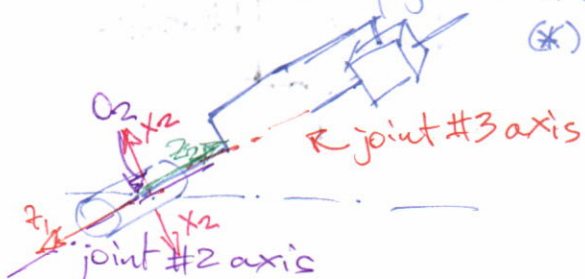


Since the common normal degenerates in a point, x_1 is only constrained to be on the normal axis to plane defined by z_0 and z_1 directions

- (-) possible choices for x_1 , both normal to z_0 and z_1

- z_2 direction is determined by joint #3 axis, while O_2 is set by the intersection of z_1 and z_2 directions. Therefore, O_2 must be in the centre of joint #1, according to D-H rules.

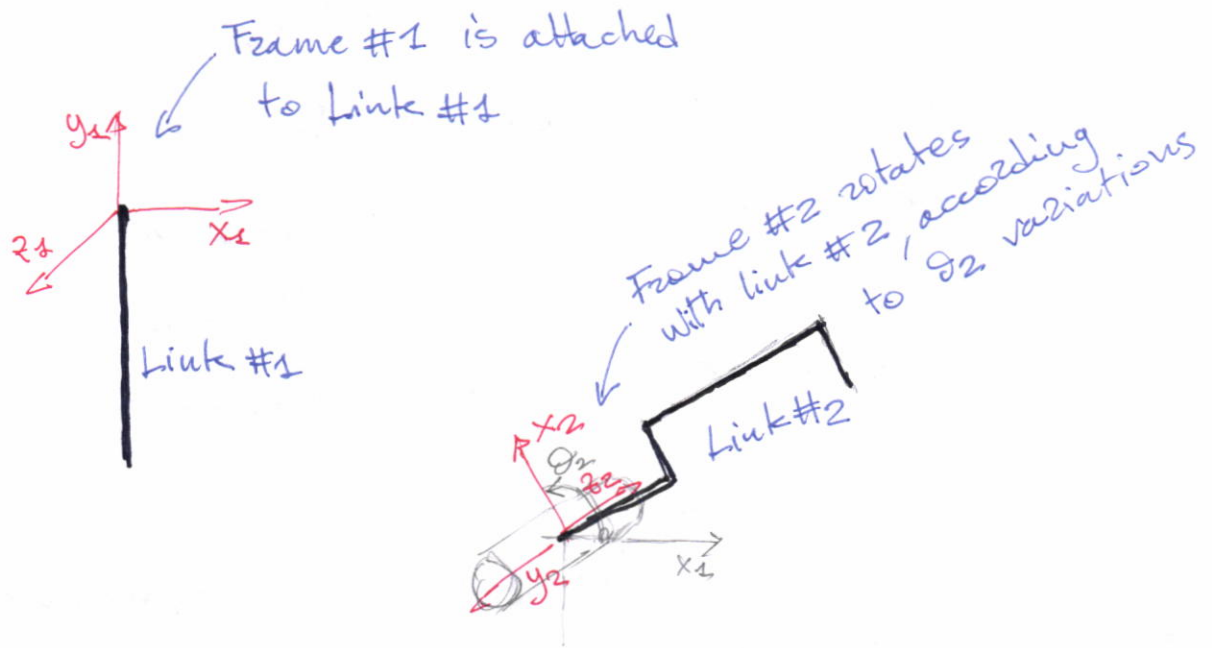
(*) The common normal degenerates in a point!



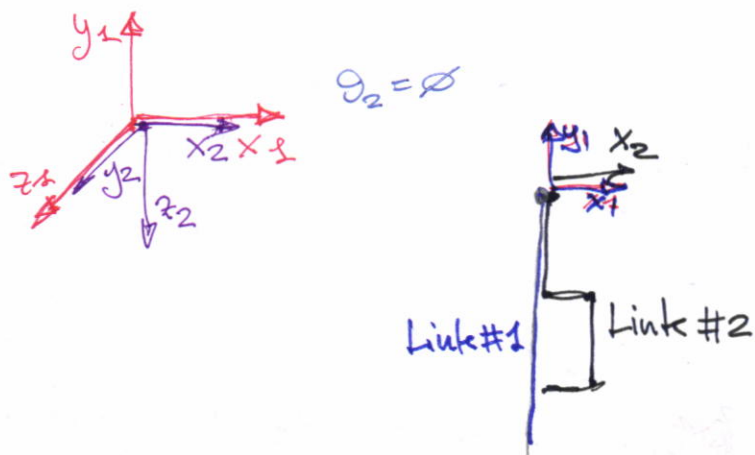
- (-) possible choices for x_2 , considering the common normal to z_1 and z_2 plane

NOTE #1

By the choice done for x_1 and x_2 , if $\theta_2 = 0$, then the manipulator is in down position. In fact:



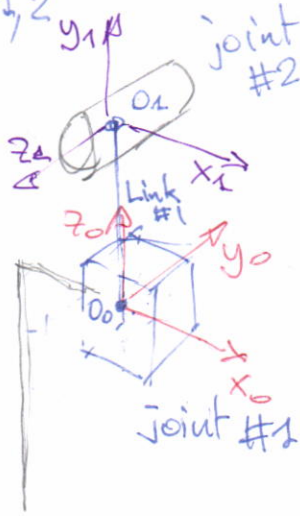
Joint variable θ_2 represents the angular position of x_2 with respect to x_1 . It comes out that if $\theta_2 = 0$, then x_1 and x_2 are aligned. Therefore, when $\theta_2 = 0$, Link #2 is in down position.



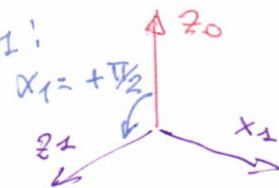
NOTE #2

(3)

Definition of the D-H parameters $a_i, \alpha_i, \theta_i, d_i$ for $i=1,2$



- a_1 defines the distance between z_0 and z_1 directions. Since the minimum distance axis degenerates in a point, $a_1 = 0$. It corresponds to the distance of O_1 from z_0 .
- α_1 defines the rotation about x_1 to align z_0 and (to) z_1 :

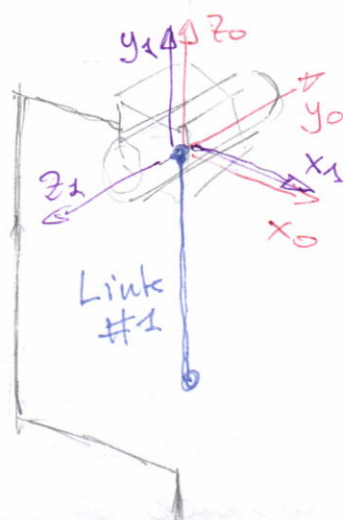
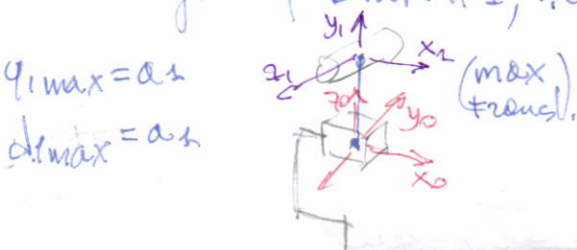


- θ_1 defines the rotation about z_0 to align x_0 and (to) x_1 . Since $x_0 \parallel x_1$, $\theta_1 = 0$
- d_1 defines the distance between x_0 and x_1 directions. It varies with joint #1 variable.

Note that:

- $O_1 - x_1 y_1 z_1$ moves according to d_1 , while $O_0 - x_0 y_0 z_0$ is a fixed frame!!!
- Considering the choices done for frames #0 and #1, when $q_1 = 0$ O_0 and O_1 overlaps, and Link #1 is completely down

- Ideally, the maximum value of d_1 is the length of Link #1, i.e. 1

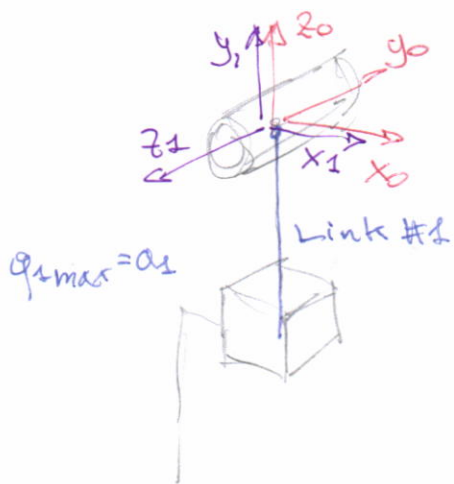


Initial position:

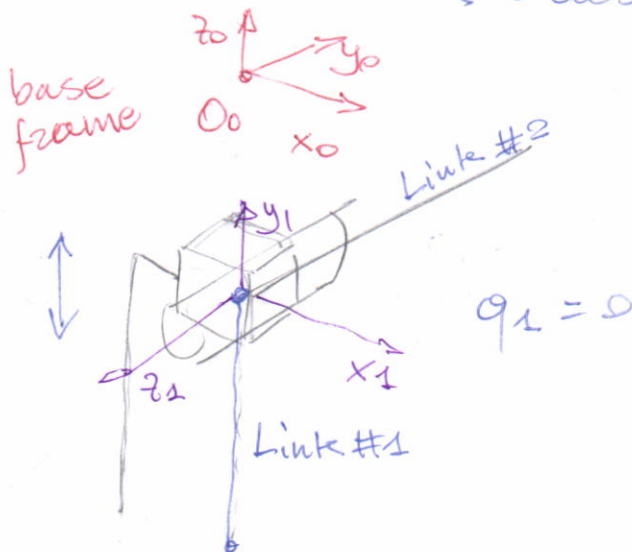
When $q_1 = 0$ (joint variable)

$d_1 = 0$ (distance between x_0 and x_1)

Please remember that d_1 varies.



This is NOT a convenient choice for frame $O_0-x_0y_0z_0$, because when $q_1=0$, the distance of x_1 from x_0 is $-a_1$ (O_0 does NOT move!!)

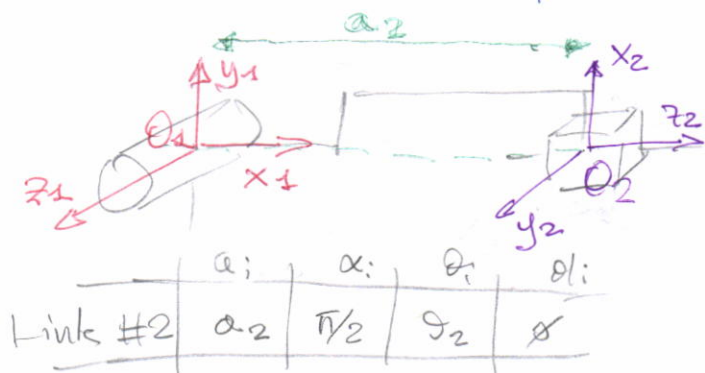


Moreover the DH parameter d_1 would be equal to $q_1 - a_1$ (or $d_1 - a_1$)

Therefore, the DH parameter varies between $-a_1$ and $d_1 - a_1$

NOTE #3

About frame #2 positioning: NOT proper choice. Why?



This choice for frame $O_2-x_2y_2z_2$ is NOT consistent with DH rules, because z_1 and z_2 directions intersect in O_1 , which is the degeneration of the minimum distance segment!! O_1 and O_2 should overlap!!

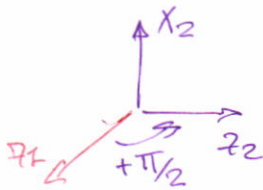
However, if frame #2 is set as in the picture, the DH parameter a_2 is NOT $\emptyset$! This is because we are not following the DH rules. Instead, a_2 is still the distance of O_2 from z_1 , equal to a_2

(5)

If we set $a_2 = 0$, the main consequence is that, when computing O_2 position by transformation matrices, we will obtain $O_2 \equiv O_1$, while O_2 moves when θ_2 changes. It can be immediately checked by looking at the last column of matrix A_2^1 , where $A_2^1(1,4)$ and $A_2^1(2,4)$ should be functions of θ_2 . If it's not, it's a big mistake!

Conversely if O_1 and O_2 coincides, position of O_2 does not change with θ_2 (see TABLE#1) as expected

Finally, z_1 must rotate of an angle $\frac{\pi}{2}$ counterclockwise (+ rotation) about x_2 axis to overlap z_2 , therefore $\alpha_2 = +\frac{\pi}{2}$



NOTE #4

About Link #3 parameters, according to the choice done for Fz.#3. When $q_3 = 0$, the (minimum) distance between x_2 and x_3 is equal to the length of Link #2, i.e. a_2 . Therefore, the d_3 parameter d_3 is equal to $a_2 + q_3$, or $a_2 + d_3$

A.2 (direct kinematic function)

TABLE #1

6

$$a_1 = a_3 = 2$$

$$a_2 = 1$$

	a_i	α_i	θ_i	d_i
Link #1	0	$\pi/2$	0	d_1
Link #2	0	$\pi/2$	θ_2	0
Link #3	0	0	0	$a_2 + d_3$

$$A_i^{i-1} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \cos \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_1^0 = \begin{bmatrix} 1 & 0 & \begin{bmatrix} z_1 \\ 0 \end{bmatrix} & \begin{bmatrix} p_1 \\ 0 \end{bmatrix} \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^1 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} s_2 \equiv \sin(\theta_2) \\ c_2 \equiv \cos(\theta_2) \end{matrix}$$

$$A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a_2 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(*) SUGGESTION: ALWAYS VERIFY THE ORIENTATION OF $O_i - x_i y_i z_i$ FRAMES BY INSPECTING EACH COLUMN OF THE TRANSFORMATION MATRICES!!!

$$T(q) = A_1^0 \cdot A_2^1 \cdot A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a_2 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow$$

$$T(q) = \underbrace{\begin{bmatrix} c_2 & 0 & \begin{bmatrix} z_2 \\ s_2 \end{bmatrix} & \begin{bmatrix} p_2 \\ 0 \end{bmatrix} \\ 0 & -1 & 0 & 0 \\ s_2 & 0 & -c_2 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{A_2^0} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a_2 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_2 & 0 & s_2 & (a_2 + d_3)s_2 \\ 0 & -1 & 0 & 0 \\ s_2 & 0 & -c_2 & -(a_2 + d_3)c_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T(q) = \begin{bmatrix} C_2 & 0 & \overset{z_3}{S_2} \\ 0 & -1 & 0 \\ S_2 & 0 & -C_2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \overset{p_3 \equiv p}{(1+d_3)S_2} \\ 0 \\ -(1+d_3)C_2 + d_1 \\ 1 \end{bmatrix} \quad (7)$$

A.3 (Geometric Jacobian)

$$J = \begin{bmatrix} z_0 & z_1(p-p_1) & z_2 \\ 0 & z_1 & 0 \end{bmatrix}$$

$$z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$z_1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$z_2 = \begin{bmatrix} S_2 \\ 0 \\ -C_2 \end{bmatrix}$$

$$p_1 = \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix}$$

$$p = \begin{bmatrix} (1+d_3)S_2 \\ 0 \\ -(1+d_3)C_2 + d_1 \end{bmatrix}$$

$$p-p_1 = \begin{bmatrix} (1+d_3)S_2 \\ 0 \\ -(1+d_3)C_2 \end{bmatrix}$$

$$z_1 \wedge (p-p_1) = \begin{bmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & -1 & 0 \\ (1+d_3)S_2 & 0 & -(1+d_3)C_2 \end{bmatrix} = (1+d_3)C_2 \underline{i} + 0 \underline{j} + (1+d_3)S_2 \underline{k}$$

$$J = \begin{bmatrix} 0 & (1+d_3)C_2 & S_2 \\ 0 & 0 & 0 \\ 1 & (1+d_3)S_2 & -C_2 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

← No translation is allowed along the y_0 axis

← No rotation is allowed about the x_0 axis

← No rotation is allowed about the z_0 axis

(8)

A.4 (position and orientation of each frame with respect to base frame)

$$q = \begin{bmatrix} 1 \\ \pi/2 \\ 0.5 \end{bmatrix} \begin{matrix} d_1 \\ \theta_2 \\ d_3 \end{matrix}$$

To be determined: R_1^0, R_2^0, R_3^0 (orientations)

$$a_2 = 1$$

$p_1, p_2, p_3 \equiv p$ (positions)

First Link frame:

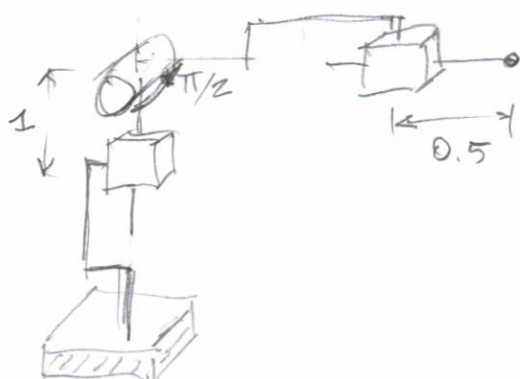
$$A_1^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1^0 \\ p_1 \end{matrix}$$

Second Link frame:

$$A_2^0 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ 0 & -1 & 0 & 0 \\ s_2 & 0 & -c_2 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_2^0 \\ p_2 \end{matrix}$$

Third Link frame:

$$A_3^0 \equiv T(q) = \begin{bmatrix} c_2 & 0 & s_2 & (a_2 + d_3)s_2 \\ 0 & -1 & 0 & 0 \\ s_2 & 0 & -c_2 & -(a_2 + d_3)c_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_3^0 \\ p_3 \end{matrix}$$

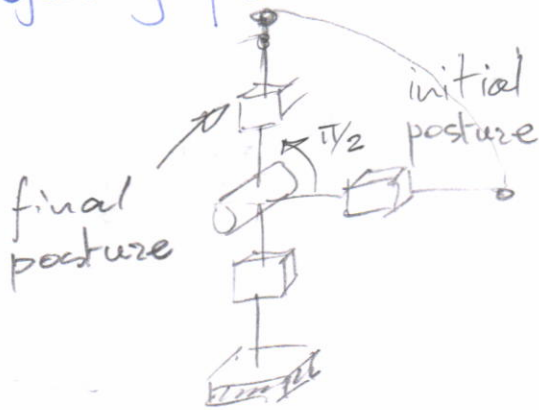


A.5 (Polynomial trajectory definition)

(9)

It is requested the end-effector frame to follow an arc of a circle in the space, of amplitude equal to $\pi/2$, starting from posture defined in A.4.

Since the rotation can be only generated by revolute joint #2, it is possible to move just this joint to obtain the desired trajectory for the end-effector frame:



Therefore, the problem reduces to defining a trajectory for joint #2, considering initial and final velocities and accelerations equal to 0, and an initial value $\theta_2^i = \pi/2$ and a final value $\theta_2^f = \pi$.

Considering that we have 6 constraints (initial/final positions, initial/final velocities, initial/final accelerations), we need a 5th order polynomial (6 parameters). Trajectory duration is requested to be of 2 seconds.

The trajectory of the joint variable is defined by: ($q_2 \equiv \theta_2$)

$$q_2(t) = a_5 t^5 + a_4 t^4 + a_3 t^3 + a_2 t^2 + a_1 t + a_0$$

under the following constraints:

$$[t_i, t_f, q_i, q_f, \dot{q}_i, \dot{q}_f, \ddot{q}_i, \ddot{q}_f] = [0, 2, \pi/2, \pi, 0, 0, 0, 0]$$

We have to impose the constraints to find a_i in $q(t)$.

(10)

By recalling that:

$$\dot{q}_2(t) = 5a_5 t^4 + 4a_4 t^3 + 3a_3 t^2 + 2a_2 t + a_1$$

$$\ddot{q}_2(t) = 20a_5 t^3 + 12a_4 t^2 + 6a_3 t + 2a_2$$

It holds:

$$\begin{array}{l|l} t=0 & \frac{\pi}{2} = a_0 \\ t=2 & \pi = a_5 \cdot 32 + a_4 \cdot 16 + a_3 \cdot 8 + \cancel{a_2 \cdot 4} + \cancel{a_1 \cdot 2} + a_0 \\ t=0 & 0 = a_1 \\ t=2 & 0 = a_5 \cdot 80 + a_4 \cdot 32 + a_3 \cdot 12 + \cancel{4a_2} + \cancel{a_1} \\ t=0 & 0 = a_2 \\ t=2 & 0 = a_5 \cdot 160 + a_4 \cdot 48 + a_3 \cdot 12 + \cancel{2a_2} \end{array} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \\ \text{velocity} \\ \\ \text{acceleration} \end{array}$$

$$\begin{array}{l} a_0 = \pi/2 \\ a_1 = 0 \\ a_2 = 0 \end{array} \left\{ \begin{array}{l} a_4 + 5a_5 = 0 \\ a_3 + 2a_4 + 4a_5 = \pi/16 \\ 3a_3 + 8a_4 + 20a_5 = 0 \end{array} \right. \left\{ \begin{array}{l} a_4 = -5a_5 \\ a_5 = 3/20 a_3 \\ a_3 = 5/8 \pi \end{array} \right.$$

$$\left\{ \begin{array}{l} a_5 = \frac{3}{32} \pi \quad (0.294) \\ a_4 = -\frac{15}{32} \pi \quad (-1.472) \\ a_3 = \frac{5}{8} \pi \quad (1.962) \end{array} \right.$$

$$q_2(t) = \frac{3}{32} \pi t^5 - \frac{15}{32} \pi t^4 + \frac{5}{8} \pi t^3 + \frac{\pi}{2}$$

$$\dot{q}_2(t) = \frac{15}{32} \pi t^4 - \frac{15}{8} \pi t^3 + \frac{15}{8} \pi t^2$$

A.6 (Find the final posture after performing the trajectory defined in A.5)

(11)

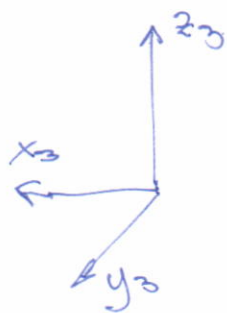
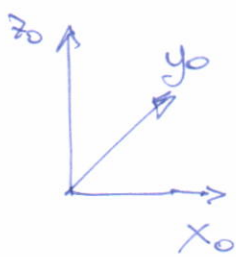
$$q_f = \begin{bmatrix} 1 \\ \pi \\ 0.5 \end{bmatrix} \quad \text{Only joint \#2 has moved!}$$

We just have to repeat computations done for A.4, by setting $\theta_2 = \pi$ instead of $\pi/2$. Only A_3^0 is requested.

$$A_3^0 = \begin{bmatrix} x_3 & y_3 & z_3 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_3 \\ 0 \\ 0 \\ 2.5 \\ 1 \end{bmatrix} \quad \text{---} \quad -(1+0.5) \times (-1) + 1$$

Manipulator is in vertical position

We now must express orientation in terms of ZYZ Euler coordinates (successive rotations about current z, y, and z axes of angles ϕ, θ, ψ , respectively)



To overlap frame #0 to frame #3 it is just needed a rotation about the z_0 axis of π or $-\pi$.

A possible solution is then $\phi = [\underbrace{\pm\pi}_{\phi}, \underbrace{0}_{\theta}, \underbrace{0}_{\psi}]$

We could even solve analytically this problem by using inversion formulas, given the R_3^0 matrix:

Since elements out of diagonal are 0, then $\theta = \arctan(\sqrt{r_{13}^2 + r_{23}^2} / r_{33}) = 0$ being r_{13} and $r_{23} = 0$. Solution degenerates, it is only possible to determine $\phi + \psi = \pm\pi$, because both rotations are about the same axis

A.7 (find velocity of each frame at time instant $t = 1$ sec, while performing trajectory defined at A.5) (12)

Firstly, it is needed to compute the joint variables and velocities at $t = 1$

Joints #1 and #3 do not move, so $q_1 \equiv d_1$ and $q_3 \equiv d_3$ are equal to 1 and 0.5. $q_2 \equiv \theta_2$ can be calculated from the trajectory equation at time $t = 1$

$$q_2(1) = \frac{3}{32}\pi - \frac{15}{32}\pi + \frac{5}{8}\pi + \frac{\pi}{2} = \frac{3}{4}\pi \Rightarrow q(1) = \begin{bmatrix} 1 \\ 3/4\pi \\ 0.5 \end{bmatrix}$$

$q(1)$ is needed to compute the Jacobian at $t = 1$ sec

[Note: $q(1)$ can be obtained by simply noticing the symmetry of $q_2(t)$ trajectory, so that at $t = 1$ sec, $q(1) = (q(0) + q(2))/2$]

As for joint velocities, $\dot{d}_1 = \dot{d}_3 = 0$, while $\dot{\theta}_2$ results from the velocity equation of the polynomial trajectory:

$$\dot{q}_2(1) = \frac{15}{32}\pi - \frac{15}{8}\pi + \frac{15}{8}\pi = \frac{15}{32}\pi \Rightarrow \dot{q}(1) = \begin{bmatrix} 0 \\ 15/32\pi \\ 0 \end{bmatrix}$$

Frame #1 is not moving, because joint #1 is not moving, therefore $v_1 = J_{L1}(q(1)) \cdot \dot{q}(1) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

For frame #2 we have (we should determine just a rotation about the y_0 axis):

$$J_{L2} = \begin{bmatrix} z_0 & z_1 \wedge (p_2 - p_1) & 0 \\ 0 & z_1 & 0 \end{bmatrix} \quad p_2 - p_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow J_{L2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$v_2 = J_{L2}(q(1)) \cdot \dot{q}(1) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ (15/32)\pi \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -15\pi/32 \\ 0 \end{bmatrix} \left. \begin{array}{l} \text{NO TRANSLATION} \\ \leftarrow \text{rotation about } y \text{ axis} \\ \text{with angular speed } -15/32\pi \end{array} \right\}$$

As for frame #3, we just have to refer to complete Jacobian J :

13

$$J = \begin{bmatrix} 0 & (1+d_3)c_2 & s_2 \\ 0 & 0 & 0 \\ 1 & (1+d_3)s_2 & -c_2 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

with $d_1 = 1$, $\theta_2 = \frac{3}{4}\pi$, $d_3 = 0.5$

$$J(q(1)) = \begin{bmatrix} 0 & -\frac{3}{2} \cdot \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ 1 & \frac{3}{2} \cdot \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{3\sqrt{2}}{4} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ 1 & \frac{3\sqrt{2}}{4} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$v_3 = J(q(1)) \cdot \dot{q}(1) = \begin{bmatrix} 0 & -\frac{3\sqrt{2}}{4} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ 1 & \frac{3\sqrt{2}}{4} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{15}{32}\pi \\ 0 \end{bmatrix} =$$

$$v_3 = \begin{bmatrix} -\frac{45\sqrt{2}}{128}\pi \\ 0 \\ \frac{45\sqrt{2}}{128}\pi \\ 0 \\ -\frac{15}{32}\pi \\ 0 \end{bmatrix} \left. \begin{array}{l} \text{Linear velocity} \\ \text{Angular velocity} \end{array} \right\}$$